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of Stavanger

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New Directions in Machine-verifiable Progress Credentials and Fully Automated and Transparent Admissions Process

Bachelor's Thesis - Electrical Engineering - April 2026

```
func (m *Manager) NewConfiguration(opts ... gorums.ConfigOption) (c *Configuration, err error) {
    if len(opts) < 1 || len(opts) > 2 {
        return nil, fmt.Errorf("wrong number of options: %d", len(opts))
    }
    c = &Configuration{}
    for _, opt := range opts {
        switch v := opt.(type) {
        case gorums.NodeListOption:
            c.Configuration, err = gorums.NewConfiguration(m.Manager, v)
            if err != nil {
                return nil, err
            }
        case QuorumSpec:
            // Must be last since v may match QuorumSpec if it is interface{}
            c.qspec = v
        default:
            return nil, fmt.Errorf("unknown option type: %v", v)
        }
    }
    // return an error if the QuorumSpec interface is not empty and no implementation
    var test interface{} = struct{}{}
    if _, empty := test.(QuorumSpec); !empty && c.qspec == nil {
        return nil, fmt.Errorf("missing required QuorumSpec")
    }
    return c, nil
}
```

I, **Hein Meling**, declare that this thesis titled, “New Directions in Machine-verifiable Progress Credentials and Fully Automated and Transparent Admissions Process” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a bachelors degree at the University of Stavanger.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.

“Programming is a nice break from thinking.”

– Leslie Lamport

Abstract

In general, the chapters described below are expected in a bachelor's thesis. Some of the listed chapters may not be applicable to your thesis, and additional chapters covering special topics may be needed. A good approach is to describe each major concept/task in a separate chapter, and describe minor related concepts in sections/subsections within the chapters.

The top-level (numbered) list corresponds to the main chapters. The second level itemization lists the contents that should be covered in that chapter. The headings typeset in **boldface** should be sections on their own. In *italic* are more detailed comments.

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Acknowledgements

I would like to thank my supervisors for their enthusiasm and help with writing this thesis.

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Chapter 1

Introduction

1.1 Questions

- Should the theory hold a different form. More history-like as several of the articles we have read?
- Should the theory be more in the shortened style as in the theory section, or more in the style of the earlier version i have added in appendix?
- Should the tables of variables also hold the values?
- Where to place the model equation?
- Are the figures ok?

1.2 Known issues/errors

- Must place citations
- Must find out how to add the sources to the document
- Full model equation is missing transport and commute
- Values for the MPC are not included, should they?
- Side-by-side figures looks small, the colors on them are also the hardest to see on a black background

- Both our documents must be combined to one, overlapping content must be removed

Page budget for Introduction: 3-5 pages.

1.3 Background and Motivation

- Awaken the reader's interest and convince her why the theme is important.
- Background information might be historical in nature, or it might refer to previous research or practical considerations.
- It should be written on a level that it's understandable by anyone with a computer science bachelor's degree.

1.4 Objectives

- Define the goals of your study.
- It might be presented as a bullet list.

1.5 Approach and Contributions

- Give a brief summary of your overall approach.
- Summarize the specific contributions that you made in this thesis (implementation, empirical results, analysis, etc.).
- It might be presented as a bullet list.

1.6 Outline

- Give an overview of the main points and the structure of your thesis.
- Examples: "Chapter 2 covers ... Chapter 3 describes ..."
- Show how the different parts (chapters) relate to each other.

Chapter 2

Theory

2.1 SIR-model

An early version of compartmental epidemic modeling, initially with a fixed population size. The populations natural birth- and death-rates are not taken into account.

S-susceptible

The part of the population that can be infected

I-infectious

The part of the population that have been infected. These can infect the susceptible population.

R-removed

The part of the population that have been removed or recovered. This part will in a SIR-model contain both recovered (healed) and dead.

The total equation is $N = S + I + R$

Where the differential equations are

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta IS}{N} \\ \frac{dI}{dt} &= \frac{\beta IS}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Here the β represents the transmission rate, the rate that someone from S is infected from someone in I.

The recovery rate γ represents the period someone is in I, and transition to R. and can be calculated by $\frac{1}{\text{days}} = \gamma$, days being the number of days someone is infected.

From the beta and gamma rates the basic reproduction ratio, R_0 , can be calculated. The number of new infections caused by one infected.

$$R_0 = \frac{\beta}{\gamma}$$

For ongoing decision-making the real-time reproduction rate is more important. We find R_t by

$$R_t = \frac{\beta_t}{\gamma_t}$$

In order to have a pandemic development, the R -value have to be larger than 1. In average, an infected person is infecting more than one new person.

2.2 SEIR

The SIR-model can be expanded to include more transmission states. A SEIR-model include the exposed-state. The ones from the susceptible population is moved to the exposed compartment when they have been exposed to someone infected. This open up the possibility to have a period where someone has been infected but is not infectious themselves.

Now $N = S + E + I + R$

With the transitions

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta SI}{N} \\ \frac{dE}{dt} &= \frac{\beta SI}{N} - \sigma E \\ \frac{dI}{dt} &= \sigma E - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Where σ is the incubation rate $\frac{1}{days}$

2.3 MPC - model predictive control

MPC is an optimized control technique, which uses a mathematical model of the process to predict the effect of adjusting the manipulated variable u . The MPC have a predictive horizon period forwards in time where it estimates how the system will behave. It uses these estimations to adjust u so that the measured output mo get close to the reference target ref . Hard strains can be implemented so that u never go past the limits of its functions, and there can be implemented hard limits to what mo is allowed to be. For every step (t), the MPC moves its horizon forward and recalculates the predicted outcome. A fundamental part of the MPC

is the cost function. The intent of the cost function is to add punishment for deviations between the measured output and the reference value, and also for making big changes in the manipulated variable. The cost function J is based on

$$J = \sum_{i=1}^N w_{mo,i} (ref_i - mo_i)^2 + \sum_{i=1}^M \omega_{u,i} w_{u,i} \Delta u_i^2$$

Where $w_{mo,i}$ is the weighting coefficient for the importance of mo , and $w_{u,i}$ is the weighting coefficient for penalizing changes in u .

In an epidemic modeling context, this can be implemented to punish high infection numbers, hospitals getting close to capacity, implementation of travel restrictions, dependent on how the MPC and variables are set up.

The MPC in MATLAB and Simulink uses a Kalman-filter in its state estimation. A Kalman-filter can be used to estimate unmeasured variables.

Chapter 3

Methods

3.1 Model development

The SIR based model was incrementally expanded by adding compartments from models used in other studies. First vaccination was added, making a transition from susceptible to vaccinated. This was followed by inclusion of breakthrough infection of vaccinated, and death, based on SIRVB and SIRVD[ref_][ref_]. The breakthrough infection allows the vaccine not being 100 % effective. To capture the impact on the healthcare system, hospitalization compartment was added. This also allows hospital load to be used as a control factor.

Finally, the model was extended to include multiple cities. Interaction between cities was implemented with a travel matrix and a commuting-based mixing model. These represent long-term relocation and short term visits between communities.

The final model is described in ??

Model Equation: The full model is described by the differential equations

$$\frac{dS}{dt} = -\lambda S - \nu S + \omega_r R + \omega_v V$$

$$\frac{dI}{dt} = \lambda S - (\gamma + \mu + h)I$$

$$\frac{dI_v}{dt} = \lambda_v V - (\gamma_v + \mu_v + h_v)I_v$$

$$\frac{dV}{dt} = \nu S - \lambda_v V - \omega_v V$$

$$\frac{dH}{dt} = hI + h_v I_v - (\gamma_h + \mu_h)H$$

$$\frac{dR}{dt} = \gamma I + \gamma_v I_v + \gamma_h H - \omega_r R$$

$$\frac{dD}{dt} = \mu I + \mu_v I_v + \mu_h H$$

Where the force of infection is defined as

$$\lambda(t) = \beta(t) \frac{I + \eta I_v}{N}$$

and

$$\lambda_v(t) = (1 - \epsilon)\lambda(t)$$

Table 3.1: Variables

Parameter	Description
β	Baseline transmission rate
λ	Force of infection
λ_v	Force of infection on vaccinated
η	Relative infectiousness of vaccinated infected
ϵ	Vaccine effectiveness
ν	Vaccination rate
γ	Recovery rate
γ_v	Recovery rate for infected vaccinated
γ_h	Recovery rate for hospitalized
μ	Death rate
μ_v	Death rate for vaccinated
μ_h	Death rate for hospitalized
h	Hospitalization rate from infected
h_v	Hospitalization rate from infected vaccinated
ω_r	Waning immunity rate for recovered
ω_v	Waning immunity rate for vaccinated

3.1.1 Macro model

Model structure

The macro model is based on a compartmental epidemic framework, with the population divided into compartments. The model was implemented in Simulink for modular construction and integration with an MPC system.

The epidemic dynamics were implemented into a subsystem representing a city. This allows expansion of more cities by duplicating the subsystem. The model is made without birth and natural death, making the population constant when dead are counted. Interaction between the cities introduced additional infection pathways.

Multi-city interaction

Interaction between cities was modeled with two mechanisms. One for long-term travel and one for commuting.

Travel between cities Long-term movement between cities was modeled using a travel matrix. The travel matrix defines the rate at which individuals move between the cities, and is applied to all the compartments except the deceased. This results in a redistribution of individuals across the cities, coupling the dynamics together.

$$\frac{dX}{dt} = XT$$

Where X is a compartment that travels, and T is the travel matrix. The travel matrix have negative outflow of the city on the diagonal, and positive inflow on the off-diagonal.

Commuting based mixing In addition to long-term travel, short-term interactions between the populations were modeled using a commuting-based approach. In the commuting mixing, individuals do not relocate to another city, but return home after being potentially being infected during the visit.

The force of infection in each city is modified based on a mixing of infected individuals:

Effective infected population was defined as $I_{eff,i} = I_i + \eta I_{v,i}$

The effective infected population was mixed by the commuting matrix $I_{mix} = M \cdot I_{eff}$

Giving the force of infection defined as $\lambda_i = \beta_i \frac{I_{mix,i}}{N_i}$

Where M is the commuting matrix describing interaction between cities. This configuration allows individuals to be exposed to infections in other cities without moving there.

MPC

Control design

A Model Predictive Controller was implemented to regulate the spread of disease by modifying the transmission rate.

The control input $u(t)$ represent the level of restrictions and is bound between 0 and 1, where $u = 0$ is no restrictions, and $u = 1$ is full restrictions.

The control input influence the transmission rate $\beta_{controlled}(t) = \beta_i(t)(1 - ku(t))$

Where k represent intervention effectiveness.

The measured output was defined as the normalized hospital load

$$mo = \frac{H1+H2}{H_{cap1}+H_{cap2}}$$

And the reference value was sat to

$$ref = H_{ref} \text{ With } H_{ref} \text{ sat to } 0.8 \text{ in the parameter file.}$$

The Simulink Model Linearizer was used to obtain a linearized model to be used as the MPC-object. The controller operates with a sample time of one day. The prediction horizon at 20, and control horizon at 10.

To ensure realistic policy implementation, the control signal u was passed through a rate limiter, limiting the rate of change of interventions. And a transport delay was added to represent the delayed effect of policy changes on the disease transmission.

The state estimator in the MPC was left to the default Kalman filter.

Chapter 4

Experimental Evaluation

4.1 Experimental Setup and Data Set

4.2 Experimental Results

4.3 Macro model behavior without control

This section presents selected simulations illustrating the behavior of the macro model without control, in order to verify that the model components produce expected epidemic dynamics.

4.3.1 Baseline epidemic dynamics

Figure 4.1 shows the epidemic dynamics of the macro model. The model consists of two interacting cities with identical initial conditions and similar parameters, connected through commuting and travel. Due to this symmetry, both cities exhibit nearly identical behavior.

For clarity, the infected compartments I_i and $I_{v,i}$ are combined and presented as the total infected population, denoted I_{i_total} . The initial conditions are given by $I(0) = 0.01$, $S(0) = 0.99$, while all other compartments are set to zero. The vaccination rates are $\nu_1 = 0.001$ and $\nu_2 = 0.002$ for *City1* and *City2*, respectively. This difference has limited influence on the epidemic dynamics in this configuration.

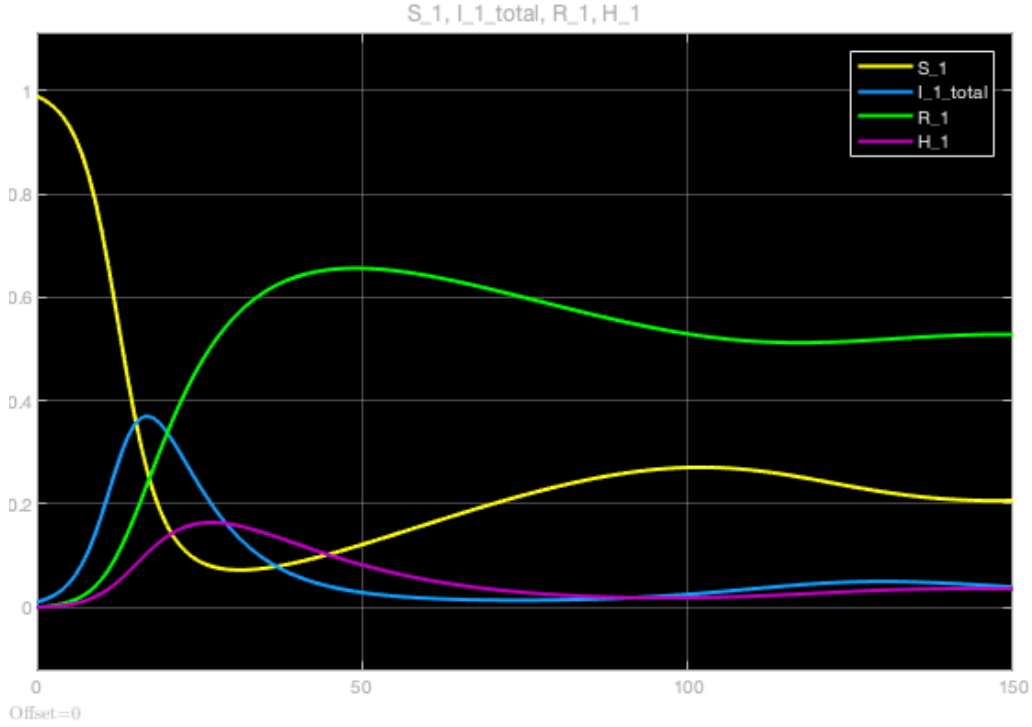


Figure 4.1: Selection of compartments from *City1*, simulated over $150T_s$

4.3.2 City to city interaction

To illustrate the effect of intercity interactions, one city is initialized with $S(0) = 1$ and all other compartments set to zero, while the other city has $S(0) = 0.99$ and $I(0) = 0.01$. Figure 4.2 shows the resulting infection dynamics in both cities under different interaction mechanisms.

The infection spreads from the initially infected city to the other city in both scenarios. When commuting is turned off, the spread is more delayed compared to the commuting only case. This demonstrates that both interaction mechanisms contribute to intercity transmission, and that the model captures the intended dynamics of epidemic spreading between cities.

4.3.3 Waning immunity

Waning immunity is included in the model to prevent the population from reaching permanent immunity and to allow for repeated epidemic waves. Without

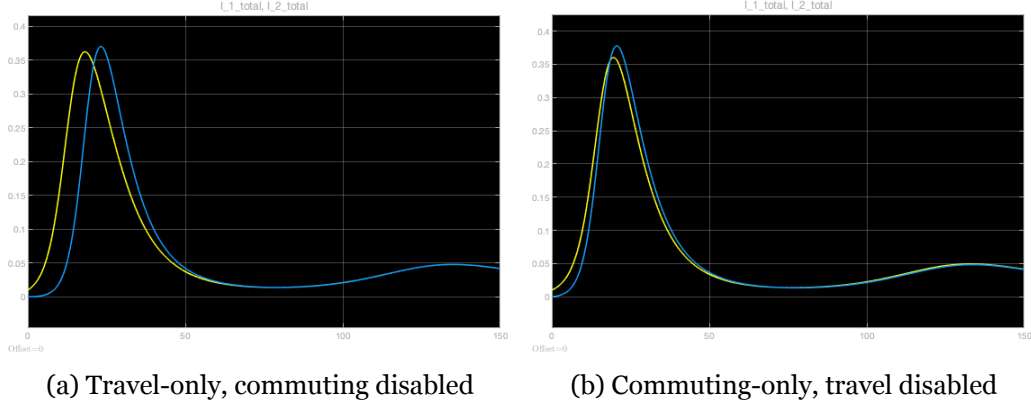


Figure 4.2: Infection dynamics in both cities under travel-only and commuting-only interaction scenarios.

waning immunity, the infection dies out after the first outbreak. Figure 4.3 shows the infection and healthcare dynamics with an extended simulation time of 500 sample times, comparing scenarios with and without waning immunity.

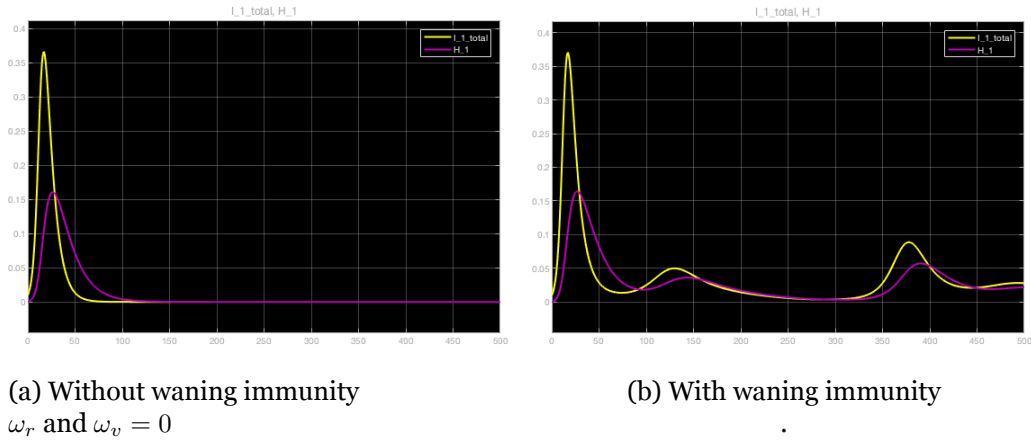


Figure 4.3: Infection and healthcare dynamics over 500 sample times, with and without waning immunity.

Without waning immunity, the infections decline to zero after approximately 60 sample times, and the healthcare demand approaches zero after around 100 sample times.

When waning immunity is included, multiple waves of infection emerge over time. This demonstrates that waning immunity significantly alters the long-term

behavior of the model and enables recurring outbreaks.

4.4 Macro model with MPC

This section presents simulation results obtained from the macro model. The purpose of the simulations is to evaluate the behavior of the model under different conditions, and the performance of the MPC controller.

The results include scenarios with and without the controller, allowing comparison between intervention and natural progression. The same initial conditions and parameter settings as in the uncontrolled case are used unless otherwise stated.

Figure 4.4 shows the development of total infections and healthcare demand across both cities without intervention.

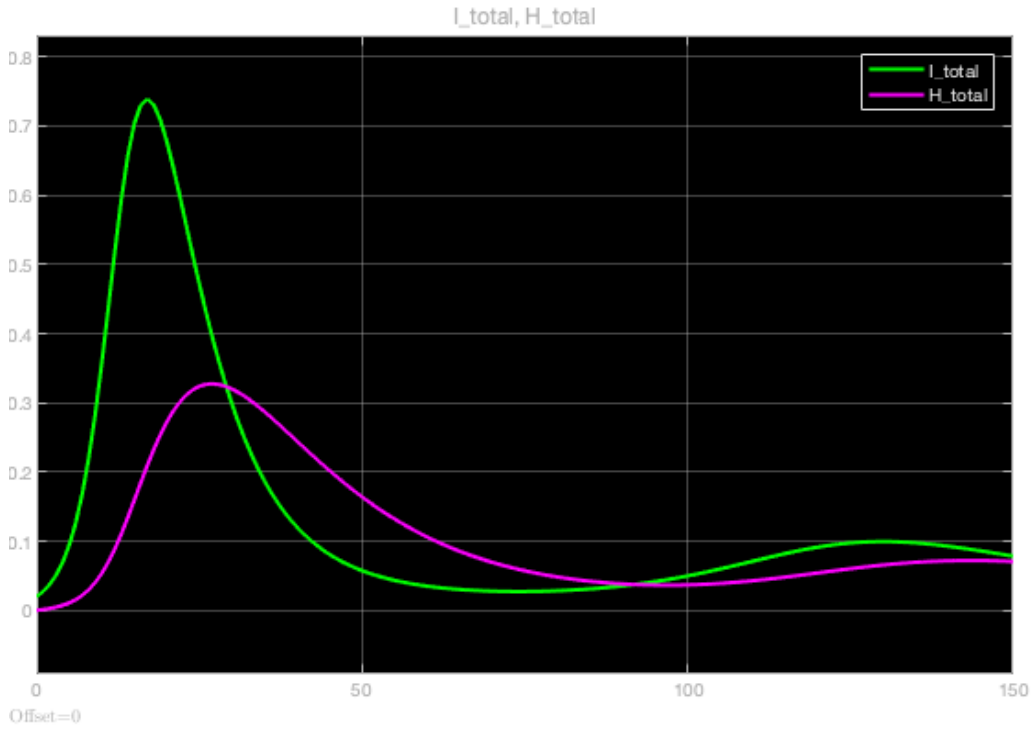


Figure 4.4: Total infections and healthcare demand across both cities without intervention.

The figure shows a rapid increase in infections, followed by a delayed increase

in healthcare demand. After reaching a peak of approximately 0.73, the infection level declines sharply before reaching a minimum and then slowly increases again. The healthcare demand follows a similar trend with a delayed peak of about 0.32, and a slower decline compared to infections.

When the MPC is implemented, both the peak values and the system dynamics change.

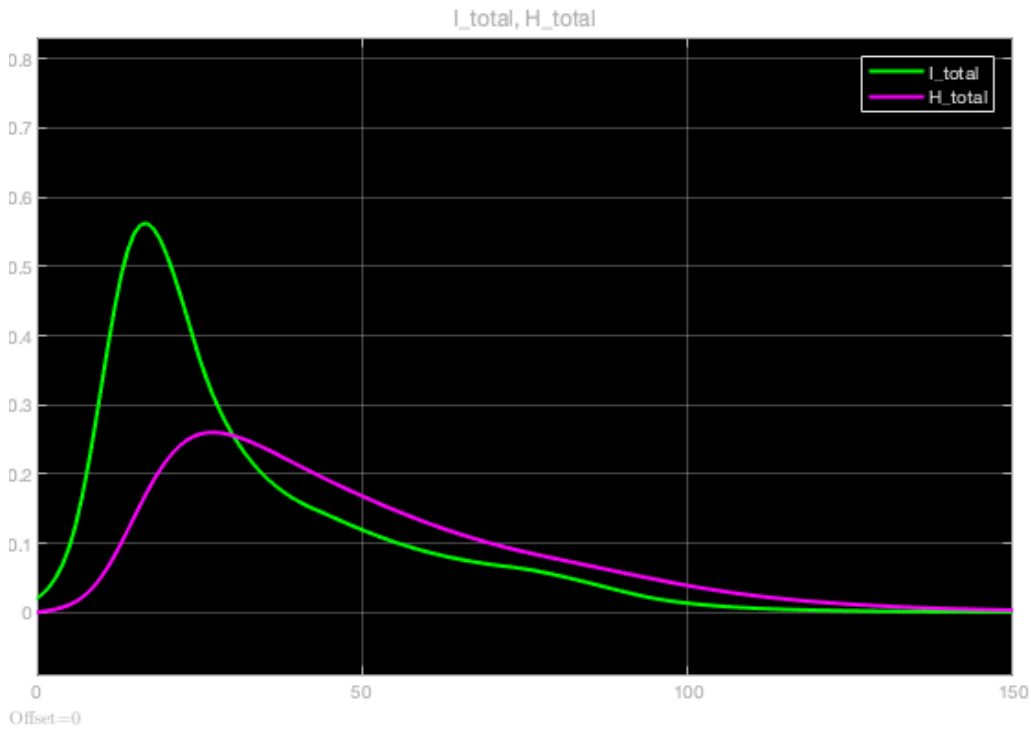


Figure 4.5: Total infections and healthcare demand across both cities with MPC intervention.

The infection peak is reduced to approximately 0.56, while the healthcare demand reaches a peak of approximately 0.26. Both exhibit a slower decline, indicating a more gradual and controlled evolution of the epidemic.

The control signal used to adjust the force of infection is shown in Figure 4.6

There is a delay of 5 sample times between the MPC and the resulting change in transmission. The delayed control signal starts early in order to suppress the healthcare demand and is reduced to 0 between the peaks of infections and healthcare demand. The control signal increases again as infections start to rise in the

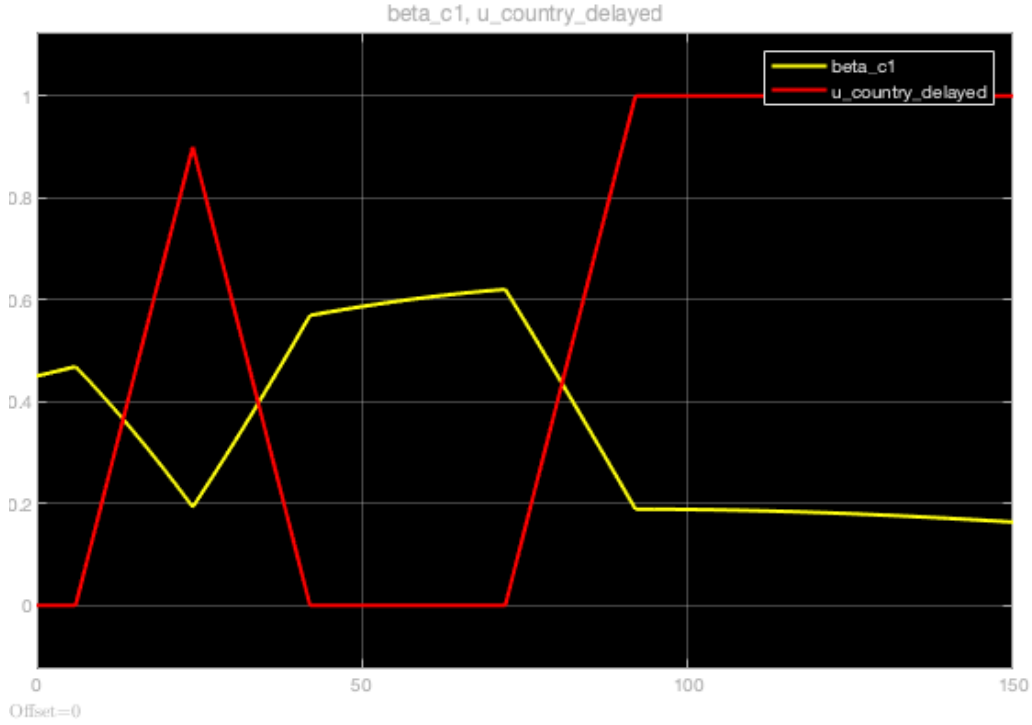


Figure 4.6: Control signal and resulting transmission rate under MPC, including delay effects.

uncontrolled scenario, and remains at its highest level for the remaining simulation time.

The combined healthcare demand is reduced from 0.32 to 0.26 and the infections reduced from 0.73 to 0.56. This demonstrates that the MPC reduces both the peak values and suppresses the development of a second wave within the simulated time frame.

When increasing the simulation time, it can be observed that the MPC suppresses the second wave.

With the simulation time set to 500, we see that the second suppression is maintained for a significantly longer period. The suppression last for about 300 days with full force, indicating that short intervention is not sufficient to prevent the second wave.

The controller behavior can be further examined by observing the measured output H_{ratio} .

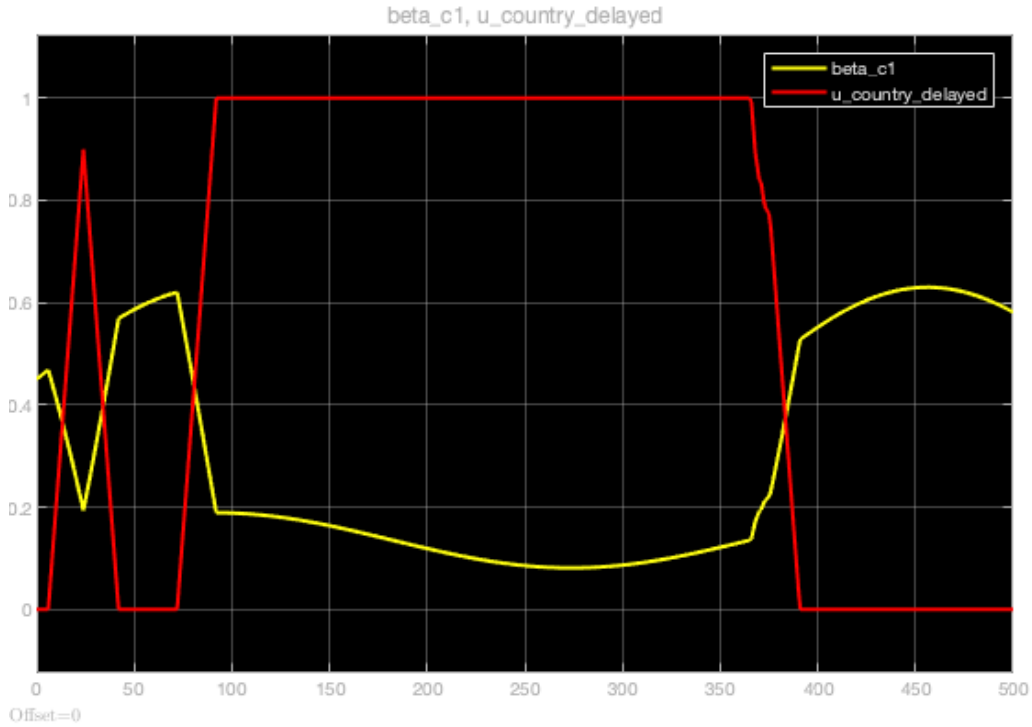


Figure 4.7: Long-term control signal behavior under MPC (simulation time 500).

The MPC continues to apply intervention even when the healthcare demand is below the reference level. This behavior indicates that the controller maintains suppression beyond what would be expected.

In the uncontrolled scenario, the healthcare system experiences significantly higher strain.

With natural progression, the healthcare demand does not go below capacity (1) until approximately 260 sample times. Then it is overloaded again after 65 sample times. These observations will be addressed in more detail in the discussion.

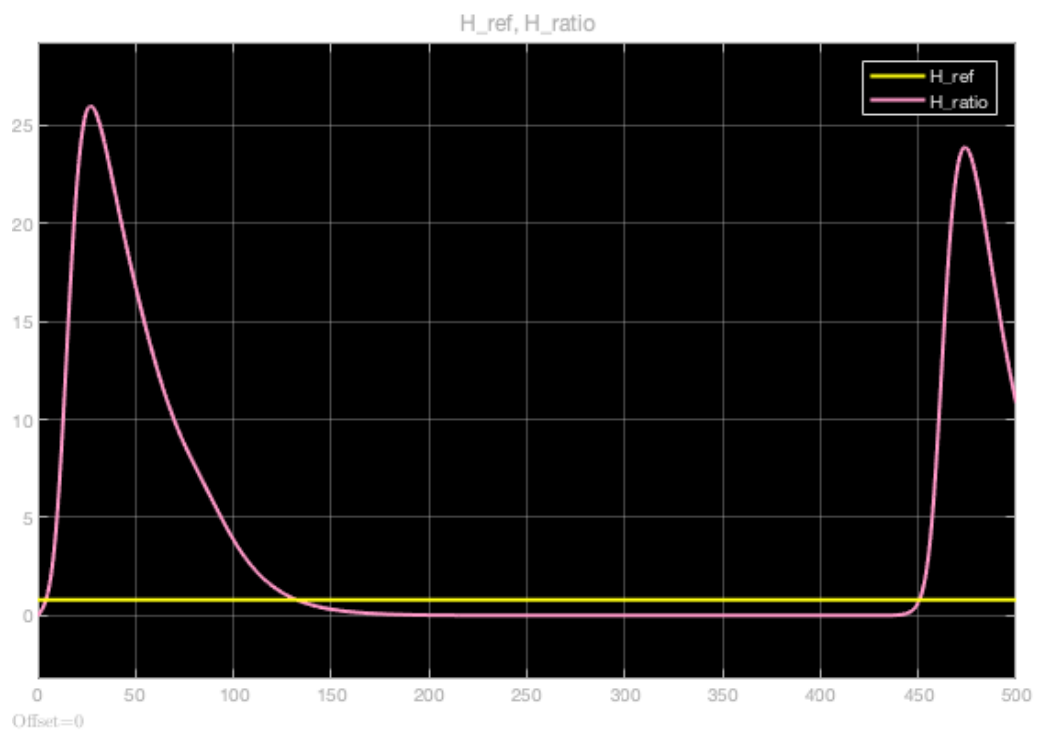


Figure 4.8: Measured output H_{ratio} compared to H_{ref} under MPC.

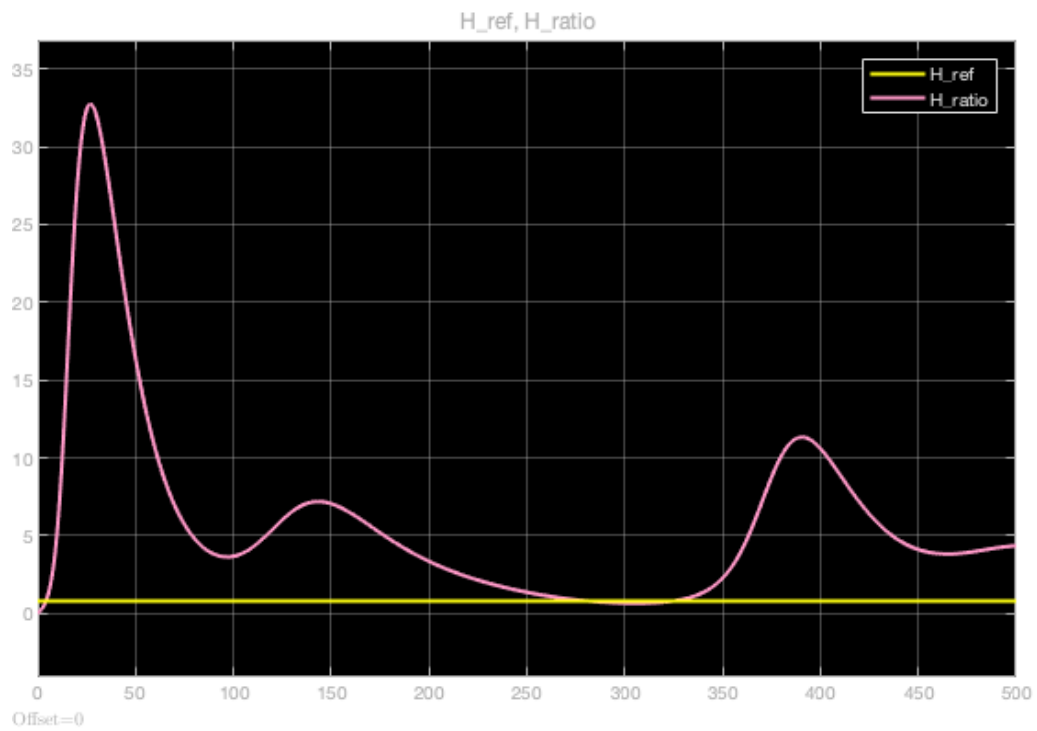


Figure 4.9: Healthcare demand ratio without control, showing prolonged over-load.

Chapter 5

Discussion

5.1 Macro model and control

This section discusses the behavior of the macro model and the performance of the implemented MPC controller. The aim is to interpret the observed results, evaluate the validity of the modeling assumptions, and highlight limitations of both the epidemic model and the control strategy. The model is not calibrated to represent a specific outbreak, but rather serves as a framework for studying epidemic dynamics and control in a multi-city setting.

5.1.1 Model assumptions and limitations

The presented macro model is based on a set of simplifying assumptions that influence the interpretation of the results. The model is not calibrated to represent a specific real-world outbreak, and most parameters are held constant. As a result, the model should be interpreted as a theoretical framework rather than a predictive tool.

The population is modeled using normalized compartment values, representing proportions of the population rather than individuals. This implies that infection levels can take small values without the epidemic dying out, which differs from stochastic models where the disease may naturally disappear at low case numbers. In this model, this behavior can be interpreted as representing undetected infections, but it also contributes to the emergence of repeated waves[1].

Additionally, the model assumes a fixed population without demographic changes

such as births, non-disease-related deaths, and age. While this simplifies the dynamics, it limits the realism of long-term simulations.

Overall, these assumptions allow for a flexible modeling framework, but they also limit the accuracy of the results and should be considered when interpreting the behavior of the model and the MPC controller.

5.1.2 Model structure and extensions

The model builds on the SIR/SEIR models that have been developed since at least 1927 [1]. Existing compartmental models with extended states and dynamics served as inspiration for the structure of the proposed model. Some of the functions were simplified, like removing age-based parameters, avoiding more complex structures such as susceptible-exposed-asymptomatic-symptomatic models.

The vaccination was simplified as a constant rate. A more realistic approach would be to add time-dependent vaccination strategies, such as batch vaccination at specific intervals. Instead a low number was used, attempting to make the vaccinated ratio of the population more realistic over a longer timeframe.

To both reduce the accumulation of vaccinated and immune through recovery, the waning immunity was introduced. This is also in accordance with diseases we see in real life, COVID-19 being a relevant example, and also the inspiration when developing the model. In the results section we saw that without waning immunity, there was not a development of following waves of infection.

The healthcare demand compartment was initially intended to be hospitalization. With the availability of hospital beds as a control output for the MPC. Because of the low reported number of hospital beds in Norway (approximately 5 beds per 100,000 inhabitants[4]) this capacity was exceeded early in the simulation. The h parameter used was based on reported hospitalization rates but may include more than just those who required hospitalization[2]. The compartment was therefore interpreted as healthcare demand rather than strictly hospitalization. In a discrete stochastic setting, a population of at least 20,000 would be required to represent a single hospital bed under these assumptions.

Both travel and commuting independently cause the cities to exhibit nearly identical compartment dynamics even when one of them starts with $I(0) = 0$. This shows that our intercity mixing is strong. This may resemble two cities in the same greater metropolitan area, such as Stavanger and Sandnes, where there

is effectively one large shared population, and some separated ones further out from the city centers. Here there also is an issue with constant parameters. The travel matrix allows up to 2% of the population to relocate every day. This represents a relatively high mobility rate for a typical urban setting but may be realistic for vacation hot-spots. Both functionalities could have been affected directly by the MPC in a more realistic approach. However, in the present formulation, the effect of the MPC on the transmission rate indirectly captures the impact of such interventions.

5.1.3 MPC behavior and control performance

In Figure 4.7, the MPC is observed to maintain intervention for approximately 300 sample times in order to suppress the second and third wave shown in Figure 4.9. This results in a prolonged period of strong control action, which may appear excessive given that the measured output falls below the reference level during this period.

The behavior is only weakly affected by the tuning parameter *Weights.ManipulatedVariables*, as similar responses were obtained for values of both 0.1 and 10. This suggests that the observed behavior is not solely due to controller tuning, but is also influenced by the underlying model and control formulation.

Several factors may contribute to this behavior. First, the choice of measured output may not fully capture the system dynamics relevant for control, and alternative outputs could potentially lead to different control actions. Second, the epidemic model is highly nonlinear, while the MPC controller is based on a linearized approximation, which may reduce prediction accuracy outside the linearization region.

Finally, the use of normalized compartment values implies that infection levels approach zero without reaching it exactly. This results in a persistent low-level infection in the model, which may cause the controller to maintain intervention in order to prevent future growth. This limitation suggests that alternative approaches, such as nonlinear or adaptive MPC controller, could potentially improve performance by better capturing the underlying system dynamics.

5.1.4 Choice of controller objective

The measured output was defined as H_{ratio} , representing the proportion of healthcare demand relative to available capacity. This choice reflects the strategy used in Norway during the later stages of the COVID-19 pandemic, where avoiding excessive strain on the healthcare system was a primary objective[3].

However, the results indicate that H_{ratio} may not be an ideal control variable in the present model formulation. The assumed healthcare capacity is relatively low, and with a hospitalization rate of $h = 0.05$, the capacity is exceeded early in the simulation as the epidemic develops. This leads to a control problem where the system operates close to or beyond capacity for much of the simulation.

Additionally, the hospitalization rate used in the model may overestimate the demand for hospital resources, as reported values can include individuals admitted for reasons other than severe illness. This introduces uncertainty in the interpretation of H_{ratio} and its suitability as a control objective.

These factors suggest that alternative or additional measured outputs, such as infection levels or combined indicators, could provide a more balanced basis for control.

5.1.5 Model applicability and future work

The presented macro model serves as a simplified framework for studying epidemic dynamics and evaluating control strategies. While the model captures key mechanisms such as vaccination, waning immunity, and intercity interactions, its level of abstraction limits its direct applicability to real-world scenarios without further calibration and validation.

To address this, an agent-based model (ABM) has been developed in parallel, representing individuals and their interactions more explicitly. The ABM can be interpreted as a more realistic representation of epidemic spread, while the macro model provides a tractable structure for control design. Comparing the behavior of these two approaches offers an opportunity to evaluate how well aggregate compartmental models capture the dynamics observed in more detailed simulations.

Furthermore, ongoing work includes the development of a model incorporating genetic algorithms (GA). Such an approach may improve the calibration of the macro model and enhance its ability to reflect real-world scenarios.

Future work could focus on integrating these modeling approaches, allowing the macro model to be continuously updated based on insights from the agent-based model and parameter optimization techniques. This would enable a more robust and informed modeling framework for decision support.

These findings highlight both the potential and the limitations of combining compartmental epidemic models with model predictive control, particularly when applied to complex and highly nonlinear systems.

Chapter 6

Conclusions

Appendix A

Instructions to Compile and Run System

Write your Appendix content here.

A.1 Alternative theory part was not finished before change - Compartmental epidemic model

SIR-model An early version of compartmental epidemic modeling, initially with a fixed population size. The populations natural birth- and death-rates are not taken into account.

S-susceptible

The part of the population that can be infected

I-infected

The part of the population that have been infected. These can infect the susceptible population.

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The part of the population that have been removed or recovered. This part will in a SIR-model contain both recovered (healed) and dead.

The total equation is $N = S + I + R$

Where the differential equations are

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta IS}{N} \\ \frac{dI}{dt} &= \frac{\beta IS}{N} - \gamma I\end{aligned}$$

$$\frac{dR}{dt} = \gamma I$$

Here the β represents the transmission rate, the rate that someone from S is infected from someone in I.

The recovery rate γ represents the period someone is in I, and transition to R. and can be calculated by $\frac{1}{days} = \gamma$, days being the number of days someone is infected.

From the beta and gamma rates the basic reproduction ratio, R_0 , can be calculated. The number of new infections caused by one infected.

$$R_0 = \frac{\beta}{\gamma}$$

For ongoing decision-making the real-time reproduction rate is more important. We find R_t by

$$R_t = \frac{\beta_t}{\gamma_t}$$

In order to have a pandemic development, the R -value have to be larger than 1. In average, an infected person is infecting more than one new person.

The SIR-model can be expanded to include more transmission states. A SEIR-model include the exposed-state. The ones from the susceptible population is moved to the exposed compartment when they have been exposed to someone infected. This open up the possibility to have a period where someone has been infected but is not infectious themselves.

$$\text{Now } N = S + E + I + R$$

With the transitions

$$\frac{dS}{dt} = -\frac{\beta IS}{N}$$

$$\frac{dE}{dt} = \frac{\beta SI}{N} - \sigma E$$

$$\frac{dI}{dt} = \sigma E - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Where σ is the incubation rate $\frac{1}{days}$

The SIR-model can be expanded to include more states. Immunity by vaccination can be included by adjusting the differential equation to

$$\frac{dS}{dt} = -\frac{\beta IS}{N} - \nu S$$

$$\frac{dI}{dt} = \frac{\beta IS}{N} - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

$$\frac{dV}{dt} = \nu S$$

The vaccination-rate ν is used to move individuals from susceptible to vaccinated V .

For a more realistic model behavior SIRVD and SIRVB models was combined to a SIRVDB-model. This includes vaccination, death, and breakthrough infection of vaccinated.

The total state equation become

$$\begin{aligned}\frac{dS}{dt} &= -\lambda S - \nu S \\ \frac{dV}{dt} &= \nu S - (1 - \epsilon)\lambda V \\ \frac{dI}{dt} &= \lambda S - (\gamma + \mu)I \\ \frac{dI_v}{dt} &= (1 - \epsilon)\lambda V - (\gamma_v + \mu_v)I_v \\ \frac{dR}{dt} &= \gamma I + \gamma_v I_v \\ \frac{dD}{dt} &= \mu I + \mu_v I_v\end{aligned}$$

Where $\lambda = \beta \frac{I + \eta I_v}{N}$ is the force of infection and η is the relative infectiousness of infected vaccinated I_v .

The breakthrough infection of vaccinated is affected by

$$\lambda_v = (1 - \epsilon)\lambda$$

With ϵ being the vaccine effectiveness against infection (1 = 100% effective, 0 = no protection from vaccine).

ν = vaccination rate (per day) γ = recovery rate μ = death rate of infected μ_v death rate for infected vaccinated γ_v recovery rate for infected vaccinated

To avoid immunity to be for life, waning immunity was introduced as ω_v and ω_r , for vaccinated and recovered.

S, V, and R adjusted to:

$$\begin{aligned}\frac{dS}{dt} &= -\lambda S - \nu S + \omega_r R + \omega_v V \\ \frac{dV}{dt} &= \nu S - \lambda_v V - \omega_v V \\ \frac{dR}{dt} &= \gamma I + \gamma_v I_v - \omega_r R\end{aligned}$$

$$\omega_v = 1/(\text{days of vaccine immunity}) \quad \omega_r = 1/(\text{days of natural immunity})$$

To make a model with populations travel between cities, a travel matrix was implemented.

$T = [-0.20.2; 0.1 - 0.1]$, meaning 20% from city1 travel to city2, and 10% from city2 travel to city1.

This matrix was applied to all states but dead, as the dead cant travel. When hospitalization was implemented, the hospitalized was not implemented to traveling either.

Multiplying the travel matrix with the states, simply adds a $+X_{travel}$ to their states. Making the cities blend their population with long travel, and permanent migration.

To implement transmissions by daily commuting, a commuting matrix was made and implemented into the force of infection.

Effective infection is calculated by

$$I_{eff,i} = I_i + \eta I_{v,i}$$

and multiplied by the commuting matrix

$$I_{mix} = M * I_{eff}$$

For a two city matrix this is

$$I_{mix,1} = I_{eff1}M_{11} + I_{eff2}M_{12} \quad I_{mix,2} = I_{eff2}M_{21} + I_{eff1}M_{22}$$

$$\lambda_i = \beta_i \frac{I_{mix,i}}{N_i}$$

Hospitalization was implemented to observe the strain on the healthcare system. (Santosh CJ, Anurag Shakya) have a SEIR model where hospitalization is called vulnerable, SEIVR. The goal is to expand the model to include those who need hospitalization. Both to improve the progression of the disease, and to predict healthcare resource impact.

The hospitalization path was implemented to the infected- and infected vaccinated pathways.

$$\frac{dI}{dt} = \lambda S - (\gamma + \mu + h)I$$

$$\frac{dI_v}{dt} = \lambda_v V - (\gamma_v + \mu_v + h_v)I_v$$

$$\frac{dH}{dt} = hI + h_v I_v - (\gamma_h + \mu_h)H$$

$$\frac{dR}{dt} = \gamma I + \gamma_v I_v + \gamma_h H - \omega_r R$$

$$\frac{dD}{dt} = \mu I + \mu_v I_v + \mu_h H$$

h = rate of hospitalization from I h_v = rate of hospitalization from I_v γ_h = recovery rate for hospitalized μ_h death rate for hospitalized



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